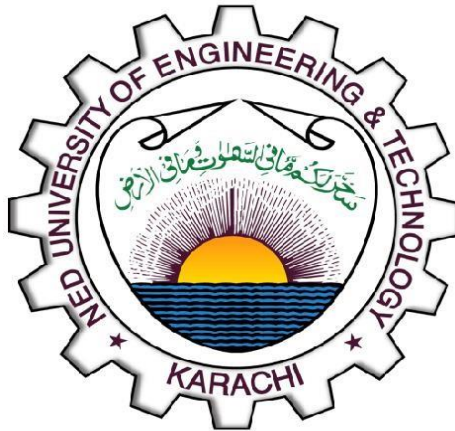


CCP PROPOSAL

PF (CT-175)



Group Members:

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Discipline: BCIT

Teacher: Sir Abdullah

1. Project Title:

Student Result Management System

2. Project Description:

This project aims to develop an efficient computerized system for generating student results. The traditional method of manually calculating results is time-consuming and prone to human errors. Our system allows users to input marks for multiple subjects, calculates percentages automatically, assigns appropriate grades based on predefined criteria, and displays results in a structured format. This eliminates manual workload and ensures accuracy and consistency in performance evaluation.

3. Project Methodology:

The system is implemented using structured programming concepts. It takes input from the user, stores it temporarily in memory using arrays, processes it to calculate total marks and percentage, and displays the final output in a tabular format. Input validation ensures that marks entered are within the acceptable range, preventing incorrect data entry.

3.1 Input / Dataset:

- Student Name is entered to identify the result report
- Number of Subjects to determine input size
- Individual Subject Names for clarity
- Marks entered must be between 0 and 100 to maintain result integrity

3.2 Tools & Technologies:

- Programming Language: C Language is used due to its efficiency and control over memory.
- Compiler: GCC or Dev-C++ is used to compile and run the program.
- Data Storage: Arrays are used to store marks and subject names.
- Operating System: Works smoothly on both Windows and Linux environments.

3.3 Algorithm:

1. Input student information and subject details.
2. Validate user input to avoid incorrect marks.
3. Loop through all subjects to sum marks.
4. Calculate percentage and assign grade using decision structures.
5. Display result in a formatted structure.

3.4 Objectives:

- Reduce manual effort in calculating results.
- Ensure accuracy through automatic calculations.
- Present results clearly for student understanding.
- Implement programming concepts including loops, functions, and arrays.

3.5 Timeline:

Day 1: Problem understanding, flowchart, and planning.

Day 2: Program coding and logic implementation.

Day 3: Testing program using different sample inputs.

Day 4: Documenting the project and preparing final submission.

3.6 Expected Outcomes:

- A working program capable of generating accurate student results.
- Improved reliability by ensuring no invalid marks is processed.
- Easy-to-understand output format beneficial for students and teachers.

3.7 Goals:

- Achieve consistent and accurate grade calculations.
- Write clean and maintainable code.
- Make the system flexible for future enhancements like GUI or database integration.

4. Justification:

This project involves logical thinking, structured problem solving, and proper application of core programming concepts. The requirement of validating input, calculating results, and formatting outputs demonstrates the application of loops, conditional statements, user-defined functions, and dynamic input handling. Such integration makes it an effective computing-level task.

5. Industrial / Commercial Scope:

This system can be scaled to support:

- School and college examination result systems
- Online result checking portals
- Automated academic record maintenance
- Integration with student management systems for large-scale educational institutions